In [16]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [17]:
```
import pandas as pd
```

In [18]:
```
df = pd.read_csv("data.csv")
```

In [19]:
```
df.head()
```

Out[19]:

	Make	Model	Year	Engine Fuel Type	Engine HP	Engine Cylinders	Transmission Type	Driven_Wheels	Nur D
0	BMW	1 Series M	2011	premium unleaded (required)	335.0	6.0	MANUAL	rear wheel drive	
1	BMW	1 Series	2011	premium unleaded (required)	300.0	6.0	MANUAL	rear wheel drive	
2	BMW	1 Series	2011	premium unleaded (required)	300.0	6.0	MANUAL	rear wheel drive	
3	BMW	1 Series	2011	premium unleaded (required)	230.0	6.0	MANUAL	rear wheel drive	
4	BMW	1 Series	2011	premium unleaded (required)	230.0	6.0	MANUAL	rear wheel drive	

In [20]:
```
df.shape
```

Out[20]:
```
(11914, 16)
```

In [21]:
```
df.columns
```

Out[21]:
```
Index(['Make', 'Model', 'Year', 'Engine Fuel Type', 'Engine HP',
       'Engine Cylinders', 'Transmission Type', 'Driven_Wheels',
       'Number of Doors', 'Market Category', 'Vehicle Size', 'Vehicle Style',
       'highway MPG', 'city mpg', 'Popularity', 'MSRP'],
      dtype='object')
```

In [22]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11914 entries, 0 to 11913
Data columns (total 16 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Make               11914 non-null  object
 1   Model              11914 non-null  object
 2   Year               11914 non-null  int64
 3   Engine Fuel Type   11911 non-null  object
 4   Engine HP          11845 non-null  float64
 5   Engine Cylinders   11884 non-null  float64
 6   Transmission Type  11914 non-null  object
 7   Driven_Wheels      11914 non-null  object
 8   Number of Doors    11908 non-null  float64
 9   Market Category    8172 non-null   object
 10  Vehicle Size       11914 non-null  object
 11  Vehicle Style      11914 non-null  object
 12  highway MPG        11914 non-null  int64
 13  city mpg           11914 non-null  int64
 14  Popularity         11914 non-null  int64
 15  MSRP               11914 non-null  int64
dtypes: float64(3), int64(5), object(8)
memory usage: 1.5+ MB
```

In [23]:
```
df.describe()
```

Out[23]:

	Year	Engine HP	Engine Cylinders	Number of Doors	highway MPG	city m
count	11914.000000	11845.00000	11884.000000	11908.000000	11914.000000	11914.0000
mean	2010.384338	249.38607	5.628829	3.436093	26.637485	19.7332
std	7.579740	109.19187	1.780559	0.881315	8.863001	8.9877
min	1990.000000	55.00000	0.000000	2.000000	12.000000	7.0000
25%	2007.000000	170.00000	4.000000	2.000000	22.000000	16.0000
50%	2015.000000	227.00000	6.000000	4.000000	26.000000	18.0000
75%	2016.000000	300.00000	6.000000	4.000000	30.000000	22.0000
max	2017.000000	1001.00000	16.000000	4.000000	354.000000	137.0000

In [24]:
```
df.isnull().sum()
```

Out[24]:

	0
Make	0
Model	0
Year	0
Engine Fuel Type	3
Engine HP	69
Engine Cylinders	30
Transmission Type	0
Driven_Wheels	0
Number of Doors	6
Market Category	3742
Vehicle Size	0
Vehicle Style	0
highway MPG	0
city mpg	0
Popularity	0
MSRP	0

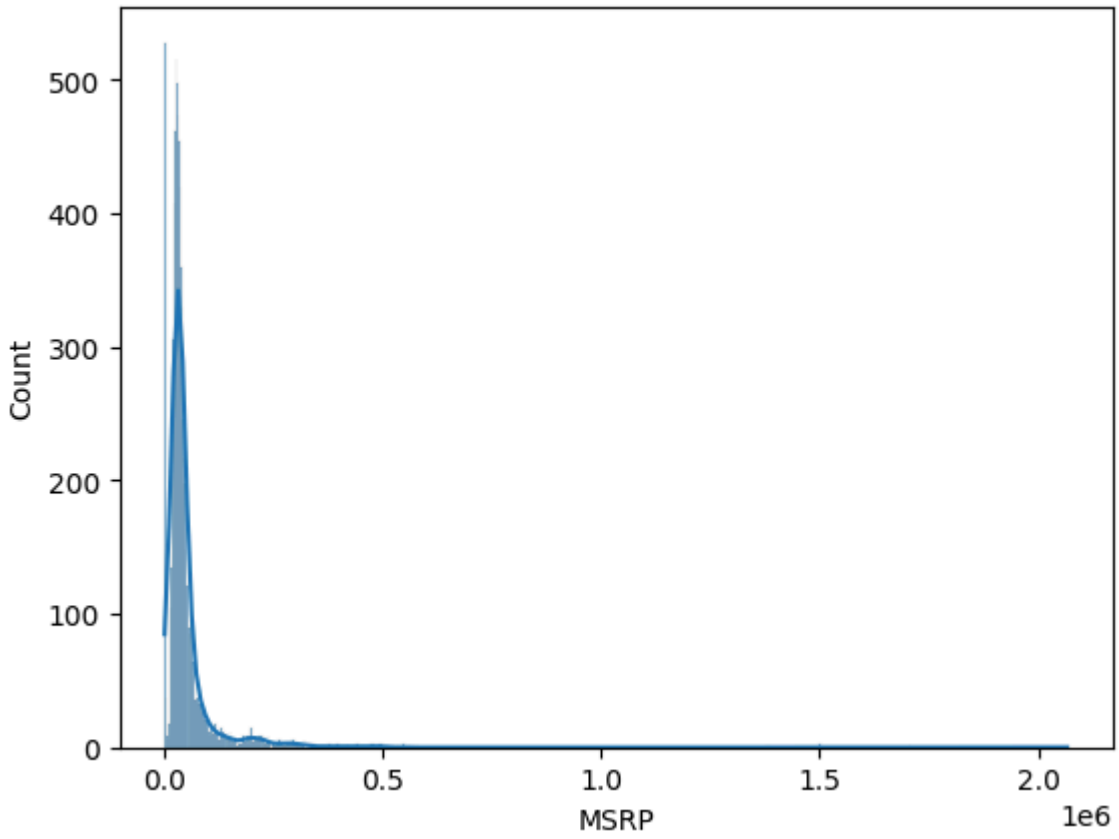
dtype: int64

In [25]:
```
df.duplicated().sum()
```

Out[25]:
```
np.int64(715)
```

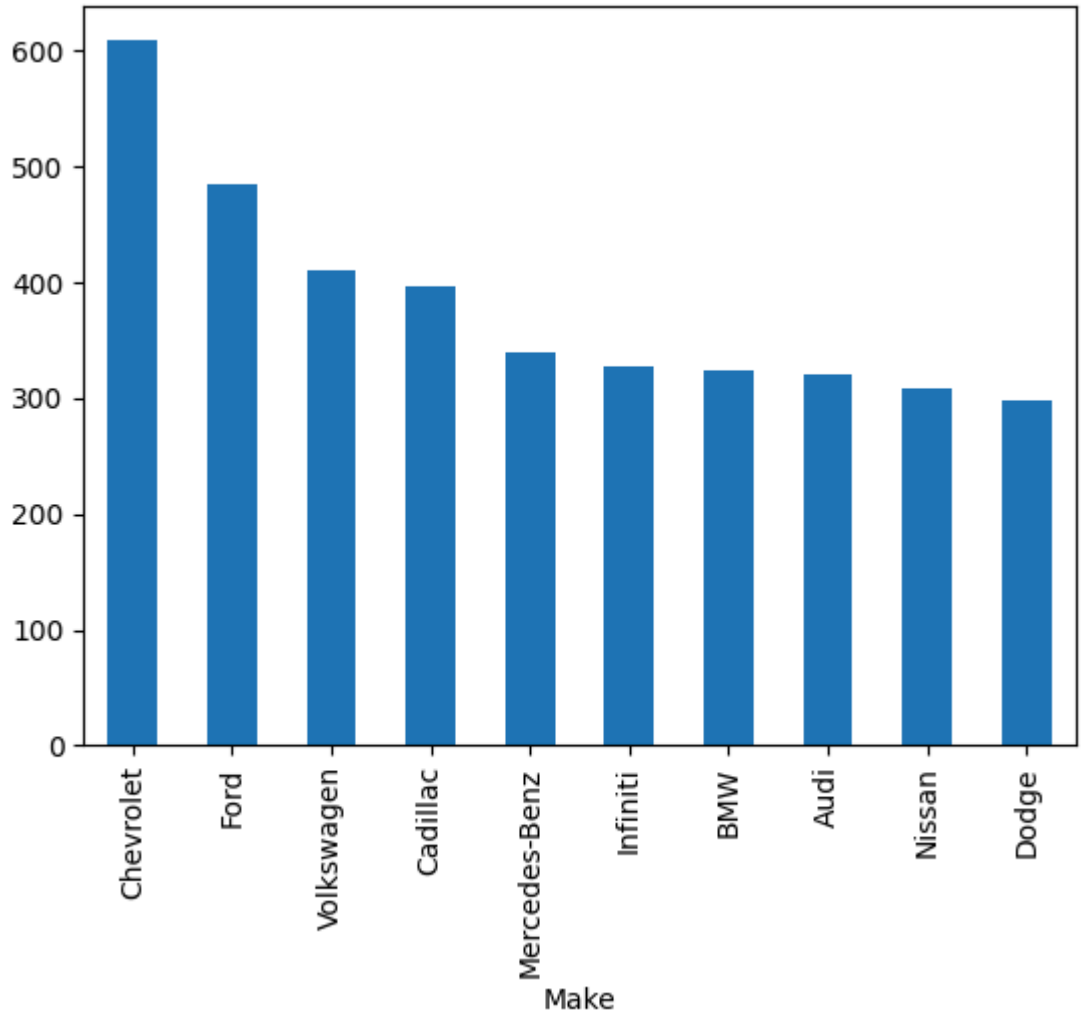
In [26]:
```
df.drop_duplicates(inplace=True)
df.dropna(inplace=True)
```

In [27]:
```
sns.histplot(df['MSRP'], kde=True)
plt.show()
```



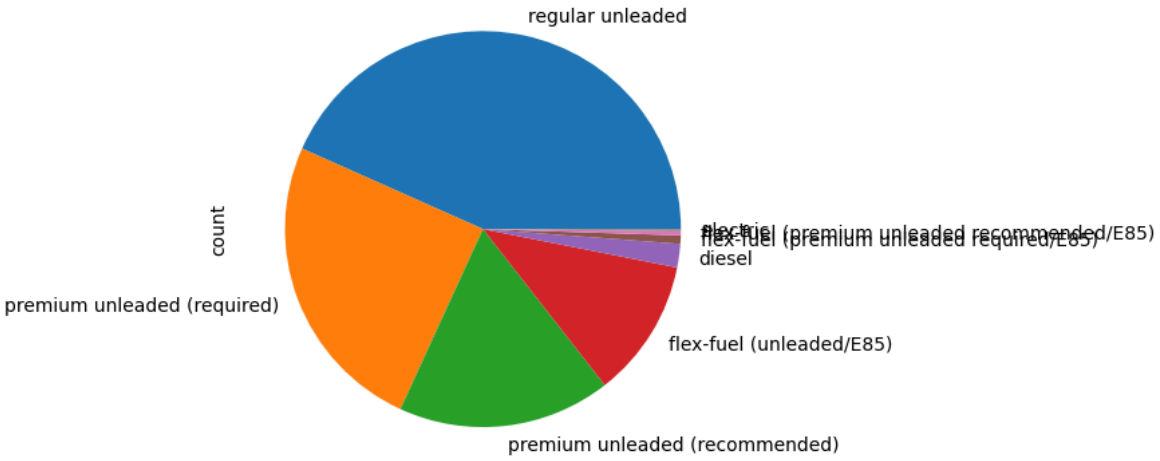
In [28]:
```
df['Make'].value_counts().head(10).plot(kind='bar')
```

Out[28]: <Axes: xlabel='Make'>



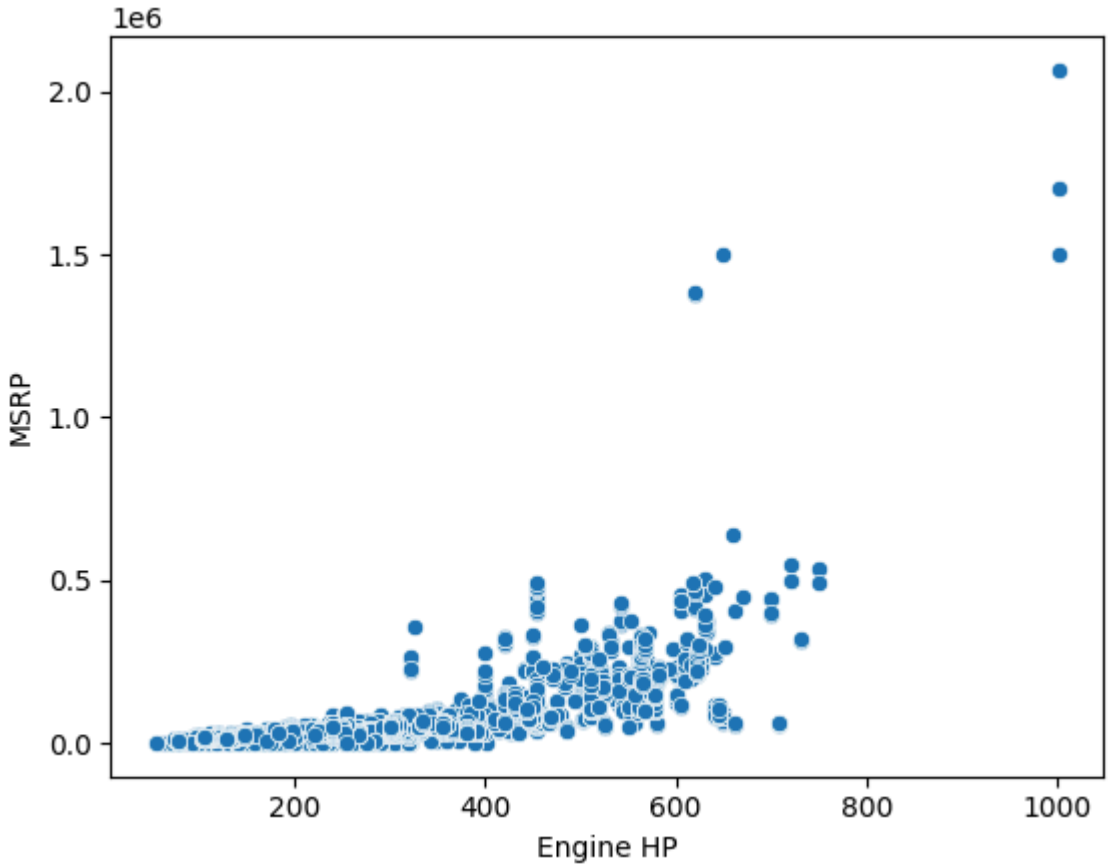
In [29]:
```
df['Engine Fuel Type'].value_counts().plot(kind='pie')
```

Out[29]: <Axes: ylabel='count'>



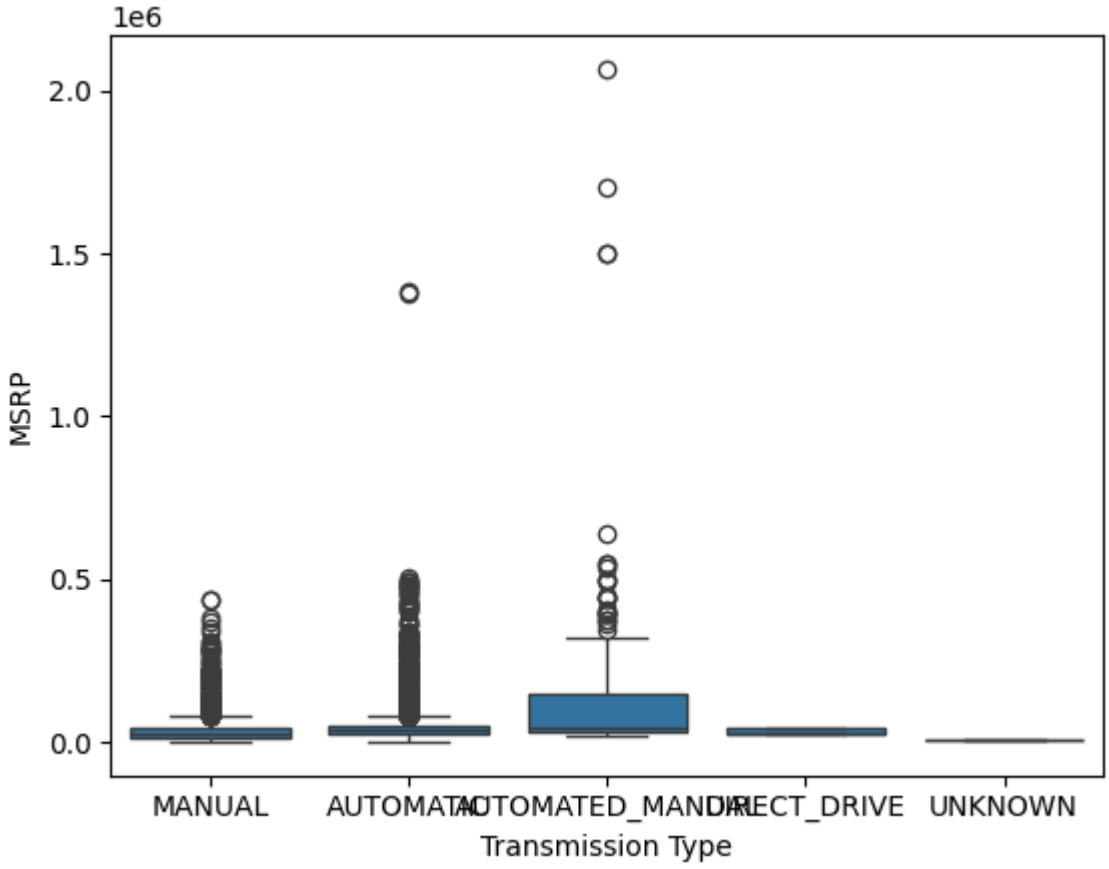
In [30]:
```
sns.scatterplot(x='Engine HP', y='MSRP', data=df)
```

Out[30]: <Axes: xlabel='Engine HP', ylabel='MSRP'>



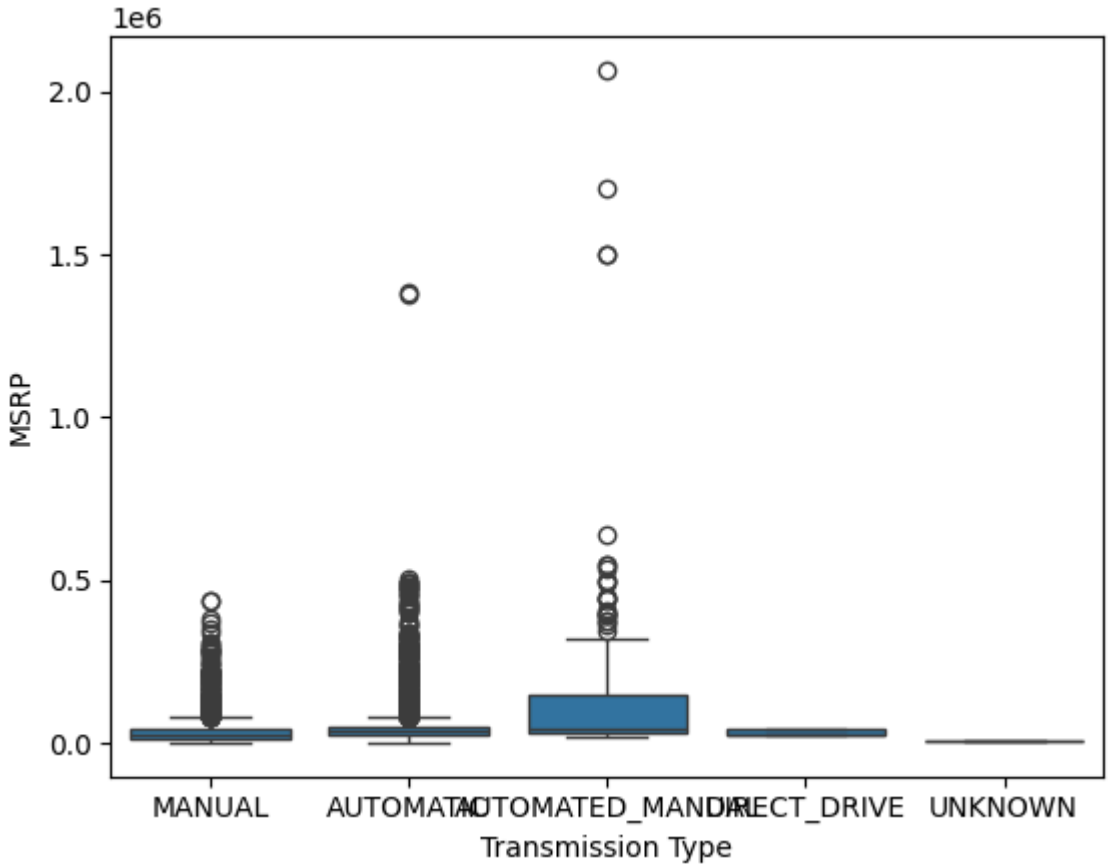
In [31]:
```
sns.boxplot(x='Transmission Type', y='MSRP', data=df)
```

Out[31]: <Axes: xlabel='Transmission Type', ylabel='MSRP'>



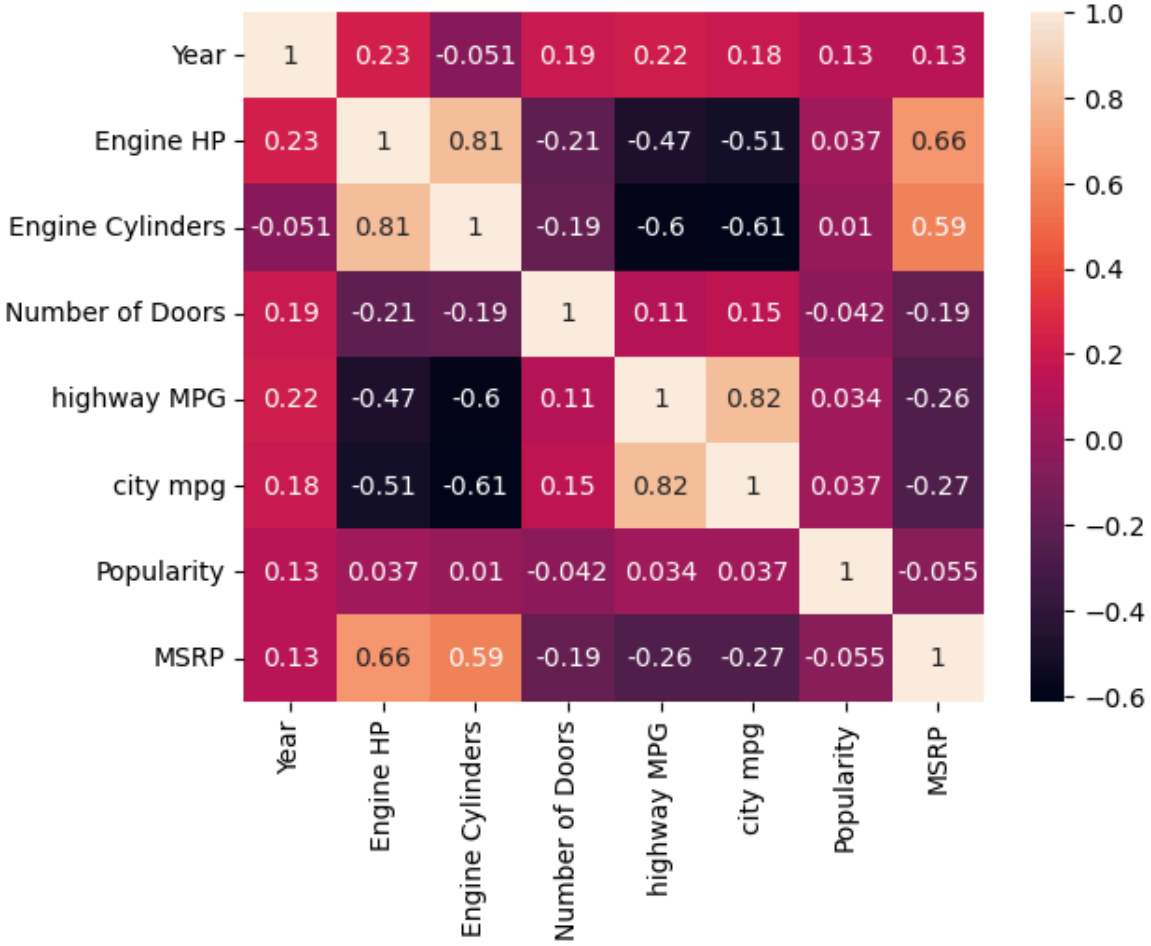
In [32]:
```
sns.boxplot(x='Transmission Type', y='MSRP', data=df)
```

Out[32]: <Axes: xlabel='Transmission Type', ylabel='MSRP'>



In [33]:
```
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

Out[33]: <Axes: >


Year
Engine HP
Engine Cylinders
Number of Doors
highway MPG
city mpg
Popularity
MSRP
1 0.23 -0.051 0.19 0.22 0.18 0.13 0.13
0.23 1 0.81 -0.21 -0.47 -0.51 0.037 0.66
-0.051 0.81 1 -0.19 -0.6 -0.61 0.01 0.59
0.19 -0.21 -0.19 1 0.11 0.15 -0.042 -0.19
0.22 -0.47 -0.6 0.11 1 0.82 0.034 -0.26
0.18 -0.51 -0.61 0.15 0.82 1 0.037 -0.27
0.13 0.037 0.01 -0.042 0.034 0.037 1 -0.055
0.13 0.66 0.59 -0.19 -0.26 -0.27 -0.055 1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6